Rudra Nil Basu

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Education

B.Tech in **Computer Science** 2014-2018
West Bengal University of Technology, Kolkata 7.88/10.0

Higher Education in **Council for the Indian School Certificate Examinations** 1999-2014
Vivekananda Mission School, Kolkata, India ISC: 94% ICSE: 92%

Experience

Tech Lead 2023 - Present
Medable Inc, Bangalore

- Architected and led the end-to-end development of Medable's event-driven, rule-engine-based clinical study-building platform, replacing a manual, script-based workflow with a self-service platform and reducing study delivery time by 80%; adopted as the standard workflow for 5–6 new studies/month.
- Built an AI-assisted debugging agent that performs initial analysis of production issues, reducing engineering effort and time-to-resolution for support tickets.
- Rearchitected the study extension development workflow from a fully manual process into an agent-driven development system, enabling agents to translate study requirements into working implementations with minimal engineer intervention
- Architected and delivered a self-service notification platform that enabled non-technical users to configure event-driven notification workflows through a UI, eliminating manual scripting and reducing engineering effort for study setup
- Led a team of 4 engineers, driving technical design, implementation, and code reviews while collaborating with Product, Sales, and CIS teams to define requirements, shape solutions, and deliver core study automation initiatives

Senior Software Engineer 2018 - 2023
HackerRank, Bangalore

- Led the end-to-end development of HackerRank's Mobile Development Assessments, delivering Android, iOS and React Native coding support across the Interviews platform while providing technical leadership to a team of two engineers spanning frontend, backend and infrastructure.
- Architected an event-driven emulator orchestration and compilation pipeline, enabling real-time synchronization of application state between interviewers and candidates while optimizing emulator lifecycle management for cost-efficient execution.
- Expanded HackerRank Interviews to support end-to-end Android, iOS and React Native coding assessments, rolling the capability out to all enterprise customers.
- Improved test case fetch time on the code-checker platform, which is the core service that powers HackerRank Community, HackerRank for Work, Interviews platform and Bing Search which in-turn reduced the overall submission execution time.
- Improved the run-time of read-only DB based submissions on the code-checker platform, where read-only DB questions constitute 90% of DB based questions.
- Maintained the service used to send out transaction and promotional emails and SMS, which were used to send emails to 300k+ community users per day while effectively tracking email opens and email link clicks.
- Build the HackerRank Jobs product and ensured effective syncing of application states between the In-Product Jobs Dashboard and all the supported ATS systems
- Build the Skills Verification Certification and Mock tests product for the Community, which has around 1 Million attempts and 800k awarded certificates.

Google Summer of Code student May 2017 - August 2017
KDE

- Proposed and added new activities and ported existing activities from GTK+ version to QT version of GCompris which are aimed at teaching children aged 2-10 years old about physics around submarines (usage of velocity and ballast tanks), digital electricity circuits (from simple gates to BCD counters and 7 segment display) and Family Trees.

Projects

Custom Game Engine **(C/C++)** 2018-Present

- Built a game engine from scratch in C/C++, using the Windows API with no third-party engine
- Implemented core engine systems including software rendering, platform layer, game loop, input handling, audio, asset loading, memory management, and debugging/profiling infrastructure
- Designed the engine around a low-level platform abstraction, with direct control over memory, rendering, timing, and system resources

Video Game Projects **(Unity3D, Blender, C#)** 2013-Current

- Developed 15+ independent video game projects, including entries for multiple game jams, covering gameplay programming, systems design, and 3D development. Available at: rudranilbasu.github.io and rudra.itch.io
- The Myth (2019): Developed a complete game in 10 days for a game jam; played 1,000+ times on itch.io, including by several prominent gaming YouTubers like Markiplier
- IGDC Recognition: 3× runner-up for Student Game of the Year for Fortior (2015), DuraTron (2016), and Evolve (2017)

Technical Skills

- C/C++, Golang, Ruby on Rails, NodeJS, TypeScript, Python, MySQL, MongoDB, Java, C#, Git, Amazon AWS, Google Cloud Platform (GCP), Docker, Kubernetes, LaTeX Unity, Blender